

Will Epperson

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 Google Scholar

Education

Ph.D. in Human Computer Interaction

August 2020 - June 2025

Carnegie Mellon University

Advisors: Dominik Moritz, Adam Perer

Thesis: Interactive Data Profiling

M.S. in Human Computer Interaction

August 2020 - May 2023

Carnegie Mellon University

Advisors: Dominik Moritz, Adam Perer

B.S. in Computer Science

August 2016 - May 2020

Georgia Institute of Technology

GPA: 4.0, Summa Cum Laude, threads in Intelligence and Modeling/Simulation

Industry Experience

Microsoft Research, Redmond, WA

August 2025 - Present

Senior Software Engineer, AI Frontiers

Research engineer working on AI agents, multi-agent systems, AI evaluations, and environments.

Microsoft Research, Redmond, WA

Summer 2024

Research Intern, AI Frontiers - HAX Group

Mentor: Gagan Bansal, Victor Dibia

Research intern working on developer tools for multi-agent AI systems.

Databricks, San Francisco, CA

Summer 2022

Software Engineering Contractor

Mentor: Kanit Wongsuphasawat

Designed and delivered production feature for creating dashboards by specifying fields of interest in a dataset.

Microsoft Research, Redmond, WA

Summer 2021

Research Intern, VIDA Group

Mentor: Steve Drucker, Rob DeLine

Research intern working on data science tools. Lead project around reuse and sharing in data science, published at ICSE 2022. Co-author on project around visualizing data frame differences published at CHI 2022.

Point72 Asset Management, New York, NY

Summer 2019

Data Analytics Intern, Market Intelligence Group

Mentor: Trevor Rempel

Worked as Data Scientist in alternative data space to clean, model, and understand large datasets

Ultimate Software, Weston, FL

Summer 2018

Software Development Intern, Innovation Strategies Team

Mentor: Joseph Cutrono

Designed and developed Slack app to integrate with the UltiPro HR management tool. App published to Slack app store.

The Home Depot, Atlanta, GA

Summer 2015

Software Development Intern

Developed web app for tracking candidate progress throughout hiring process for internal HR use.

Research Experience

Carnegie Mellon University, Pittsburgh, PA

August 2020 - June 2025

Graduate Researcher, Data Interaction Group (DIG)

Advisor: Dominik Moritz, Adam Perer

Member of the DIG research group, working on novel data visualizations, ML interpretation techniques, and interactive data systems.

Georgia Institute of Technology, Atlanta, GA

January 2019 - May 2020

Undergraduate Researcher, Polo Club of Data Science

Advisor: Duen Horng (Polo) Chau

Member of the Polo Club of Data Science working on novel data visualizations to find fairness issues in Machine Learning models

Georgia Institute of Technology, Atlanta, GA

January 2018 - May 2019

Undergraduate Researcher, Automated Algorithm Design

Advisor: Jason Zutty, Greg Rohling

Worked on EMAD algorithm design engine to implement sentiment analysis pipeline to analyze news articles to aid in predicting stock price movements using genetic algorithms. Led project to visualize the genetic algorithm evolution process.

Publications

SocialReasoning-Bench: Measuring Whether AI Agents Act in Users' Best Interests

Tyler Payne, Will Epperson, Safoora Yousefi, Zachary Huang, Gagan Bansal, Wenyue Hua, Maya Murad, Asli Celikyilmaz, Saleema Amershi

A benchmark for measuring AI agents' negotiation ability in calendar coordination and marketplace settings.

Microsoft Research Blog. 2026.

[🔗 Project](#) [📄 Blog](#) [🔗 Code](#)

Whimsical Strategies Break AI Agents: Generating Out-of-Distribution Adversarial Strategies at Scale

Zachary Huang, Tyler Payne, Gagan Bansal, Will Epperson, Wenyue Hua, Adam Fourney, Amanda Swearngin, Maya Murad, Ece Kamar, Saleema Amershi

Automatically generating out-of-distribution adversarial strategies that reliably manipulate frontier agents in negotiations.

Microsoft Research Blog. 2026.

[🔗 Project](#) [📄 Blog](#)

Red-Teaming a Network of Agents: Understanding What Breaks When AI Agents Interact at Scale

Gagan Bansal, Shujaat Mirza, Keegan Hines, Will Epperson, Zachary Huang, Whitney Maxwell, Pete Bryan, Tyler Payne, Adam Fourney, Amanda Swearngin, Wenyue Hua, Tori Westerhoff, Amanda Minnich, Maya Murad, Ece Kamar, Ram Shankar Siva Kumar, Saleema Amershi

We evaluated a live network of over 100 always on AI agents and found new failure modes that only emerge at the network level.

Microsoft Research Blog. 2026.

[🔗 Project](#) [📄 Blog](#)

Magentic Marketplace: An Open-Source Environment for Studying Agentic Markets

Gagan Bansal, Wenyue Hua, Zezhou Huang, Adam Fourney, Amanda Swearngin, Will Epperson, Tyler Payne, Jake M. Hofman, Brendan Lucier, Chinmay Singh, Markus Mobius, Akshay Nambi, Archana Yadav, Kevin Gao, David M. Rothschild, Aleksandrs Slivkins, Daniel G. Goldstein, Hussein Mozannar, Nicole Immorlica, Maya Murad, Matthew Vogel, Subbarao Kambhampati, Eric Horvitz, Saleema Amershi

An open-source simulation environment for studying how LLM agents behave as buyers and sellers in two-sided markets.

Preprint. 2025.

[🔗 Project](#) [📄 PDF](#) [🔗 Code](#)

Interactive Data Profiling

Will Epperson

My PhD thesis on interactive systems to help users automatically understand their data while programming.

Carnegie Mellon University (CMU). Pittsburgh, PA, 2025.

[🔗 Project](#) [📄 PDF](#)

Texture: Structured Exploration of Text Datasets

Will Epperson, Arpit Mathur, Dominik Moritz, Adam Perer

Texture is a general purpose text exploration tool with interactions for using LLMs to derive data from text.

Open source. 2025.

[🔗 Project](#) [📄 PDF](#) [🔗 Code](#)

Interactive Debugging and Steering of Multi-Agent AI Systems

Will Epperson, Gagan Bansal, Victor Dibia, Adam Fourney, Jack Gerrits, Erkang Zhu, Saleema Amershi

AGDebugger is an interactive debugging tool for multi-agent AI systems.

2025 CHI Conference on Human Factors in Computing Systems (CHI). Yokohama, Japan, 2025.

[🔗 Project](#) [📄 PDF](#) [🔗 Code](#)

Over-Relying on Reliance: Towards Realistic Evaluations of AI-Based Clinical Decision Support

Venkatesh Sivaraman, Katelyn Morrison, Will Epperson, Adam Perer

We discuss how evaluations of human-AI decision support systems can move beyond reliance as the primary metric.

2025 CHI Workshop on HCI+Health (CHI). Yokohama, Japan, 2025.

[🔗 Project](#) [📄 PDF](#)

Guided Statistical Workflows with Interactive Explanations and Assumption Checking

Yuqi Zhang, Adam Perer, Will Epperson

GuidedStats is a Jupyter extension that helps data scientists perform statistical analyses with guided workflows.

VIS 24: IEEE Conference on Data Visualization (VIS). St Pete Beach, Florida, 2024.

[🔗 Project](#) [📄 PDF](#) [🔗 Code](#)

Dead or Alive: Continuous Data Profiling for Interactive Data Science

Will Epperson, Vaishnavi Gorantla, Dominik Moritz, Adam Perer

AutoProfiler is a Jupyter extension that helps data scientists understand their data and find issues during analysis through continuous data profiling.

VIS 23: IEEE Conference on Data Visualization (VIS). Melbourne, Australia, 2023.

[🔗 Project](#) [📄 PDF](#) [🔗 Code](#) 🏆 **Best Paper Honorable Mention**

A Declarative Specification for Authoring Metrics Dashboards

Will Epperson, Kanit Wongsuphasawat, Allison Whilden, Fan Du, Justin Talbot

Quick dashboarding presents a novel specification for dashboard authoring, comprised of sections of metrics combined with dimensions.

VDS at VIS 23: Visual Data Science Symposium (VDS). Melbourne, Australia, 2023.

[🔗 Project](#) [📄 PDF](#) 🏆 **Best Paper**

Leveraging Analysis History for Improved In Situ Visualization Recommendation

Will Epperson, Doris Jung-Lin Lee, Leijie Wang, Kunal Agarwal, Aditya Parameswaran, Dominik Moritz, Adam Perer

Solas is a visualization recommendation tool that uses the history of analysis for in situ recommendations in Jupyter.

EuroVis 22: Eurographics Conference on Visualization (EuroVis). Rome, Italy, 2022.

[🔗 Project](#) [📄 PDF](#) [🔗 Code](#)

Strategies for Reuse and Sharing among Data Scientists in Software Teams

Will Epperson, April Yi Wang, Robert DeLine, Steven M. Drucker

Interviews and a survey with 149 data scientists at Microsoft revealed five distinct strategies for sharing and reusing analysis code along with factors that encourage or discourage reuse.

ICSE 22: ACM International Conference on Software Engineering (ICSE). Pittsburgh, PA, 2022.

[🔗 Project](#) [📄 PDF](#) 🎥 **Recording** 📄 **Slides**

Diff in the Loop: Supporting Data Comparison in Exploratory Data Analysis

April Yi Wang, Will Epperson, Robert DeLine, Steven M. Drucker

Diff in the Loop supports tracking, comparing, and visualizing differences in datasets during iterative data analysis.

SIGCHI 22: ACM Symposium on Computer Human Interaction (CHI). New Orleans, LA, 2022.

[🔗 Project](#) [📄 PDF](#)

RECAST: Interactive Auditing of Automatic Toxicity Detection Models

Austin P. Wright, Omar Shaikh, Haekyu Park, Will Epperson, Muhammed Ahmed, Stephane Pinel, Diyi Yang, Duen Horng (Polo) Chau

Interactive Auditing of Automatic Toxicity Detection Models

24th ACM Conference on Computer-Supported Cooperative Work & Social Computing. 2021.

[🔗 Project](#) [📄 PDF](#)

FairVis: Visual Analytics for Discovering Intersectional Bias in Machine Learning

Angel Cabrera, Will Epperson, Fred Hohman, Minsuk Kahng, Jamie Morgenstern, Duen Horng (Polo) Chau

Discovering intersectional ML Bias through interactive visualization.

IEEE Conference on Visual Analytics Science and Technology (VAST). Vancouver, Canada, 2019.

[🔗 Project](#) [▶ Demo](#) [📄 PDF](#) [📖 Blog](#) [🎥 Recording](#) [</> Code](#)

Talks

Interactive Data Profiling for Python DataFrames with AutoProfiler and Texture

PyCON 2024

May 2024

Dead or Alive: Continuous Data Profiling for Interactive Data Science

VIS 23: IEEE Visualization Conference

October 2023

A Declarative Specification for Authoring Metrics Dashboards

VDS at VIS 23: Visual Data Science Symposium

October 2023

Leveraging Analysis History for Improved In Situ Visualization Recommendation

EuroVis 22: Eurographics Conference on Visualization

June 2022

Strategies for Reuse and Sharing among Data Scientists in Software Teams

ICSE 22: ACM International Conference on Software Engineering

May 2022

FairVis

VIS 19: IEEE Visualization Conference

October 2019

Honors and Awards

PURA: President's Undergraduate Research Award

2019

\$1500 research grant to continue work on FairVis project

Stamps President's Scholarship

2016

Full ride scholarship given to 40 incoming freshman at Georgia Tech

Mentees

During my PhD, I have had the pleasure of mentoring the following undergraduate and masters students on research projects.

Leijie Wang

Summer 2021 - Fall 2021

Visualization recommendation for python in notebooks using history

Asad Sheikh

Fall 2021 - Spring 2022

Visualization recommendation for SQL using history

Vaishnavi Gorantla

Spring 2022 - Spring 2023

Fact generation from data and presentation as text

Yuqi Zhang

Spring 2023 - Spring 2024

Notebook extension for guided statistical analysis

Allie Feldman

Summer 2023 - Spring 2024

Guided Data Analysis

Teaching

Graduate Teaching Assistant

Spring 2023

Carnegie Mellon University, Pittsburgh, PA

Programmable User Interfaces, Instructor: Scott Hudson

Taught recitation, designed assignments for class about UI design and intro HTML, CSS, & Javascript.

Graduate Teaching Assistant

Fall 2022

Carnegie Mellon University, Pittsburgh, PA

Interactive Data Science, Instructor: Adam Perer and John Stamper

Graded and office hours for class about using jupyter, visualization, steamlit and related tech for data science.

Undergraduate Teaching Assistant

August 2017 - December 2018

Georgia Institute of Technology, Atlanta, GA

Intro to Database Systems (CS 4400), Instructor: Monica Sweat

Designed projects, held office hours and graded for relational databases class.

Service

Reviewer for CHI 2025, VIS 2024, VIS 2023, CHI 2023, VIS 2022, CHI 2022, CSCW 2021, VIS 2021.

Leadership & Activities

Student Ambassador

January 2019 - May 2020

Georgia Institute of Technology Alumni Association

Serve as official representative of the Institute at events/tours for alumni, prospective students, and special guests.

Executive Board Member -- Threads Co-chair

January 2018 - May 2019

Stamps Scholars National Convention 2019

Executive board member of Stamps Scholars National Convention, a 3-day conference with over 700 student attendees. Responsible for 20-person committee that plans and coordinates the different content threads of the convention.

Skills

Programing Languages: Python (Advanced), Javascript/Typescript (Advanced), SQL (Intermediate)

Toolkits, Frameworks, Software: Svelte, React, Pytorch, Scikit-learn, Git, VegaLite, D3

Natural Languages: English (Native), Spanish (Advanced)